BUSINESS CASE COMPETITION 2026

SOCRATIC AI: Transforming Gemini & Gemini Notebook from Task Shortcuts to Human-Centered Learning Companions

Elevating Brand Impact Through an Innovative-Driven Strategy in Indonesian Higher Education

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Proposal Title: SOCRATIC AI Framework for Higher Education (Semifinal Proposal)
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SOCRATIC AI: Transforming Gemini & Gemini Notebook from Task Shortcuts to Human-Centered Learning Companions in Indonesian Higher Education

1. Executive Summary

Indonesia's roughly 9 million higher-education students already use AI, but mostly as a shortcut: a faster way to finish a task rather than a better way to understand it. While 84.5% of surveyed pre-service teachers have used AI tools, only 44.8% have received formal training (Oktaviani et al., 2026), and about 51% of students worry that AI erodes critical thinking (Agusnaya & Nirmala, 2025). For Google this is a brand and market risk as well as an educational one. Occasional, task-based usage is easily substituted by any competing chatbot and builds neither loyalty nor institutional trust, while universities increasingly view generative AI through an academic-integrity and governance lens (Susanto et al., 2026).

SOCRATIC AI repositions Gemini and Gemini Notebook as the *thinking partner* of Indonesian students, one that asks before it answers when learning is at stake. It rests on three pillars: (1) the **3C Socratic Framework** (Challenge, Check, Construct), delivered through Bahasa Indonesia prompt libraries, Notebook templates and Classroom workflows; (2) a **three-tier go-to-market** that combines institutional integration through Google Workspace for Education, where Gemini and Gemini Notebook are already included free of charge (Google for Education, n.d.-a; n.d.-b), with telco-enabled access for 3T regions and 500 Google Student Champions; and (3) a **US$1.5 million, three-year roadmap** with phase gates, risk controls and a transparent ROI model.

Base-case Year-3 outcome: 2.5 million activated users across 300+ institutions, of whom 1.0 million are sustained users; more than 60% of Notebook sessions grounded in uploaded sources; at least 15% of users from 3T regions; 15,000 trained educators; a programme cost of US$0.60 per activated user; and payback in about 19 months once paid conversion begins. Table 1 maps this proposal to the four key questions of the case.

Table 1. Proposal map to the case key questions

Key question	Our answer in one line	Section
Market analysis	Enter through first- and second-year learners (habit formation), build credibility with deep researchers via Notebook, and scale through STEM and business majors.	3
Adoption engagement	Zero-cost institutional distribution, an AARRR funnel tied to the academic calendar, and peer-led advocacy by 500 Student Champions.	5
Product relevance	The 3C framework with adaptive friction, Bahasa Indonesia templates, Classroom integration and a low-bandwidth mode.	4
Financial and responsible impact	A US$1.5M phased budget, scenario-based ROI, explicit risk mitigation and a sustainability plan beyond the fund.	6, 7

Why now (added evidence). Field evidence shows that the design of an AI tool, not only access to it, determines whether students learn. In a randomized experiment with nearly 1,000 high-school students, learners who used a standard chatbot interface scored 17% lower on a later unaided exam than students who

never had access, while a version with learning guardrails largely removed that harm (Bastani et al., 2025). At the same time, OpenAI and Google launched guided study modes in mid-2025 (Glasp, 2026), so the "guided learning" category is converging. Google's defensible advantage in Indonesia therefore lies in what generic modes do not yet offer: Bahasa Indonesia academic content, lecturer-level workflow in Classroom, process evidence for assessment, and equitable reach into 3T regions. Sections 2.5, 3.5 and 4.6 develop this evidence.

2. Problem Understanding

2.1 Context and business objective

Indonesia's education ecosystem comprises more than 45 million K-12 students and about 9 million higher-education students in over 4,500 institutions (Kemdiktisaintek, 2024). AI is used in three areas: educator support, student learning and productivity, and academic research. Google's opportunity is to move Gemini and Gemini Notebook beyond occasional, task-based use towards intentional, consistent integration in students' academic journeys, *enhancing rather than replacing* learning. We translate this into three business objectives: **O1 Adoption depth** (habitual rather than occasional use), **O2 Brand trust** (becoming the AI that universities endorse), and **O3 Responsible impact** (measurable learning and equity outcomes at sustainable cost).

2.2 Root-cause diagnosis

We separate visible symptoms from underlying causes using an issue tree and a Five-Whys test (Table 2).

Table 2. Symptom-to-root-cause analysis

Symptom	Root cause	Evidence	Implication
AI used as a shortcut, not a companion	Default experiences reward fast answers; no localized, guided learning pathway; assessments reward the finished artefact, not the reasoning	84.5% of pre-service teachers use AI; 44.8% trained (Oktaviani et al., 2026); 51% of students fear weaker critical thinking (Agusnaya & Nirmala, 2025)	Redesign the interaction (3C), not just awareness
Unequal access to cloud AI	Connectivity gap	Internet penetration 83.64% in Java vs 67.6% in 3T areas, a 16.04-point gap (Septiani, 2024)	Low-bandwidth mode and zero-rating
Educators cannot evaluate or guide AI use	Training deficit; no shared assessment practice	Usage outpaces training by 39.7 points (Oktaviani et al., 2026)	Train-the-Trainer; assessment redesign kit
Institutions hesitate to endorse AI	Governance gaps (privacy, surveillance, transparency) and tight budgets	Concerns across 56 reviewed studies (Susanto et al., 2026); BOPTN allocation of Rp6.018 trillion cut by about 50% in budget efficiency (kumparanNEWS, 2025)	Governance kit; zero-licensing-cost model

Five-Whys summary. Why do students use AI as a shortcut? Because the fastest route to a grade is a finished answer. Why? Because assessments reward the artefact rather than the thinking. Why does the tool not counter this? Because default experiences are generic, English-first and unguided. Why do educators not correct it? They are untrained and lack policy support.

Why is policy missing? Governance and budgets have lagged adoption. **The root cause is therefore a misalignment of design and incentives across the product, educator and institution layers, not a lack of AI access alone.** Solving only one layer (for example, a student awareness campaign) would leave the shortcut incentive intact.

2.3 Stakeholder needs

Table 3. Stakeholder map

Stakeholder	Core need	Main concern	What SOCRATIC AI offers
Students	Understand material, pass exams, build employable skills	Being falsely accused of AI misuse; falling behind peers	Hint-ladder tutoring; Learning Journal as proof of process; Bahasa Indonesia support
Educators	Credible assessment; less administrative load	Ungrounded AI submissions; no training	Source-anchored work, Classroom-assigned Guided Notebooks, training and policy templates
University leaders	Compliance and quality at no extra licence cost	Privacy, reputation, BOPTN constraints	Zero-licence integration, governance kit, enterprise-grade data protection
Government	Alignment with National AI Strategy 2020–2045; equity	Digital divide widening	Ring-fenced 3T budget; national learning index
Telcos	New education-linked traffic and brand value	Unclear commercial return	Zero-rated educational domains with measurable usage data
Google	Habitual use, brand trust, long-term Workspace and Gemini loyalty	Reputational risk from academic misuse	A defensible “AI that teaches thinking” position

2.4 Problem statement

How might Google make Gemini and Gemini Notebook the habitual, trusted learning companion of Indonesian higher-education students, without replacing their thinking, at zero licensing cost to universities and with equitable reach into 3T regions?

2.5 Why now: global evidence and policy anchors

Three bodies of evidence support the problem definition above and the design choices in Section 4 (Table 3a). Together they show that (i) unguided AI can reduce learning, (ii) well-designed guided AI can improve it, and (iii) policy bodies expect human-centred, privacy-safe and teacher-supported deployment.

Table 3a. Evidence base for the problem and the proposed design

Source	Key finding	Implication for SOCRATIC AI
Bastani et al. (2025)	Field experiment with nearly 1,000 high-school mathematics students. A standard GPT-4 interface raised practice performance by 48% but lowered later unaided exam grades by 17% compared with no access. A tutor-style version with hints and safeguards raised practice performance by 127% and largely avoided the exam penalty.	Unguided AI can help students perform better today and learn less. Guardrails are a product-design decision, which supports adaptive friction and the hint ladder (Section 4.1).
Kestin et al. (2025)	Randomized crossover trial with 194 Harvard physics students. An AI tutor built on pedagogical best practice produced significantly larger learning gains in less time,	Carefully designed AI tutoring can raise learning and engagement. It is best positioned as a complement to

	with higher engagement and motivation, than an in-class active-learning session.	lecturers, which supports lecturer-assigned Guided Notebooks.
UNESCO (2023)	First global guidance on generative AI in education. It calls for a human-centred approach that protects human agency, equity and linguistic diversity, and recommends regulation, data-privacy protection and teacher training.	Human-in-the-loop design, privacy by design, educator training and Bahasa Indonesia localization map directly to these recommendations (Section 4.5).
Glasp (2026); Pearson International Schools (2026)	ChatGPT Study Mode (July 2025), Gemini Guided Learning (August 2025) and Claude Learning Mode now use guided questioning instead of direct answers. Educators note that such modes help only if students engage with them actively.	Guided mode is becoming a baseline feature. Value shifts to localization, institutional workflow and evidence of student process.

Transferability caveat. These studies were run in a high-school mathematics setting (Bastani et al., 2025) and in a single Harvard physics course (Kestin et al., 2025). Their transfer to Indonesian higher education is an assumption, which the Year-1 pilot is designed to test (Section 6.1 and Section 7.3).

3. Key Analysis

3.1 Market sizing

We size the opportunity as a funnel (Figure 1). The **total addressable market (TAM)** is the 9.0 million higher-education students (Kemdiktisaintek, 2024). The **serviceable market (SAM)** is estimated at 5.4 million (60%): students in institutions that can be enrolled in Workspace for Education (or the free Google for Education Fundamentals programme) and that have usable connectivity. The **obtainable market (SOM)** is our Year-3 target of 2.5 million activated users, equal to 28% of TAM and 46% of SAM. The SAM share is a planning assumption to be validated against Google's Workspace for Education enrolment data in the first quarter (Appendix B). We also model a conservative case (1.2 million) and an optimistic case (3.5 million) in Section 6.3.

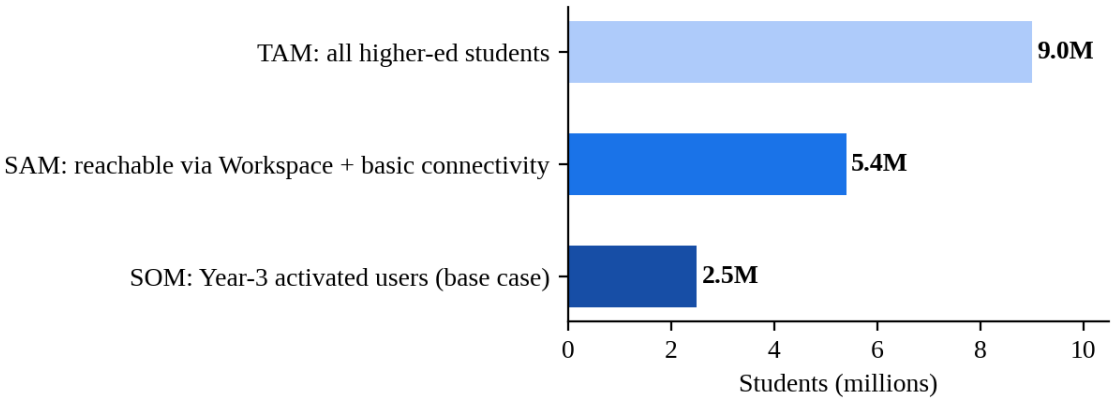


Figure 1. Market funnel: TAM, SAM and SOM (millions of students)

Source: TAM from Kemdiktisaintek (2024); SAM and SOM are team estimates.

3.2 Segmentation and prioritization

We retain three behavioural personas: **Segment A, Deep Researchers** (25%, about 2.25 million; graduate and senior undergraduates needing literature synthesis, mainly using Gemini Notebook); **Segment B, Practical Innovators** (40%, about 3.6 million; STEM, vocational and

business majors needing analysis, code and modelling, mainly using the Gemini app); and **Segment C, Foundational Learners** (35%, about 3.15 million; first- and second-year students needing conceptual clarity and exam preparation, mainly using embedded Socratic prompts). The shares are team estimates (Appendix B). Because the case asks for solutions that *enhance rather than replace* learning, we prioritize by weighted criteria rather than by size alone (Table 4).

Table 4. Segment prioritization (scores 1–5 are team judgement)

Criterion (weight)	A	B	C	Rationale
Segment size (20%)	2	5	4	B is largest
Pain intensity: shortcut reliance (25%)	4	3	5	C forms study habits
Fit with "enhance, not replace" (25%)	5	3	5	Notebook grounding; Socratic prompts
Ease of reach (15%)	3	4	4	C reachable via orientation and Classroom
Lifetime and brand value (15%)	4	3	5	Habits formed early persist
Weighted score (max 5.0)	**3.70**	**3.55**	**4.65**	

Sequencing: Segment C is the *entry wedge* (habit formation at the start of the academic journey); Segment A provides *depth and credibility* through Notebook's source-grounded research; Segment B provides *volume* through data, coding and modelling use cases once the Socratic habit is established.

3.3 Competitive position and SWOT

General-purpose AI assistants and citation-based search tools compete mainly on answer speed and are consumer-led. Few are natively embedded in the identity, storage and coursework layer that universities already administer. Gemini and Gemini Notebook are core services within Workspace for Education, covered by its terms, with data that is not human-reviewed or used to train AI models (Google Workspace Admin Help, n.d.), and Notebook answers are grounded only in the sources the user provides (Google for Education, n.d.-b). The strategic implication is to compete on *trust and institutional fit* rather than on answer speed (Table 5).

Table 5. SWOT summary

Strengths / Opportunities	Weaknesses / Threats
S: Free for institutions via Workspace; existing Classroom, Docs and Drive footprint; source-grounded Notebook; enterprise-grade data protection **O:** BOPTN-driven demand for zero-cost tools; National AI Strategy alignment; rising demand for AI literacy; academic-calendar rhythms for habit building	**W:** Generic English-first defaults; limited Indonesian academic templates; adoption depends on admins enabling features; possible overlap with existing guided-learning features **T:** Substitution by faster, friction-free chatbots; faculty resistance; privacy incidents; widening 3T divide if equity is not funded

3.4 Key insights

1. **The problem is design and incentives, not access alone.** Free access without guided use reinforces the shortcut.

2. **Trust is the most defensible position.** If universities endorse Gemini as the AI that teaches thinking, switching costs rise beyond feature parity.
3. **Zero-cost distribution turns a constraint into an advantage.** The BOPTN cut (kumparanNEWS, 2025) disadvantages paid tools but not Workspace-embedded ones.
4. **Habits are early and seasonal.** Orientation weeks, mid-terms, finals and thesis seasons are the natural activation moments.
5. **Equity needs its own budget line.** Without ring-fenced funding, adoption will concentrate in Java, where connectivity is 16.04 points higher (Septiani, 2024).
6. **Guided learning is converging across vendors, so differentiation must be local and institutional.** ChatGPT, Claude and Gemini all now offer guided study modes (Glasp, 2026); the open space is Bahasa Indonesia content, lecturer workflow, process evidence and 3T access (Table 5a).

3.5 Competitor benchmarking: guided-learning modes

Table 5a compares the three guided-learning modes publicly described in 2026 with the layer SOCRATIC AI proposes. It relies on public descriptions and secondary reviews, and should be refreshed with Google's product team before the Year-1 build scope is fixed.

Table 5a. Guided-learning modes versus the SOCRATIC AI layer

Dimension	ChatGPT Study Mode	Claude Learning Mode	Gemini Guided Learning	SOCRATIC AI layer (proposed)
Core approach	Adaptive, conversational questioning; flexible chat-based flow (Glasp, 2026)	Most strictly Socratic; resists giving direct answers (Glasp, 2026)	Structured, visual guidance with quizzes inside the Gemini interface; presents more than it asks (Glasp, 2026)	Challenge–Check–Construct with an adjustable hint ladder, on top of Gemini and Notebook
Institutional hook	Free on every plan (Glasp, 2026)	Claude for Education with Canvas integration (Glasp, 2026)	Included in Workspace for Education (Google for Education, n.d.-a)	Classroom-assigned Guided Notebooks; Learning Journal; aggregated lecturer indicators
Localization and equity (team assessment)	No Indonesia-specific academic templates or low-bandwidth mode found in sources reviewed	No Indonesia-specific academic templates or low-bandwidth mode found in sources reviewed	No Indonesia-specific academic templates or low-bandwidth mode found in sources reviewed	60+ Bahasa Indonesia templates; low-bandwidth mode; telco zero-rating; ring-fenced 3T budget

Implication. Competing on a feature race with better-funded generic modes is not necessary. Gemini already has the strongest institutional position through Workspace; SOCRATIC AI adds what turns that position into trust: local content, lecturer-visible process evidence and equitable access.

3.6 External environment scan (PESTEL)

Table 5b. PESTEL summary

Factor	Key observations	Implication
Political	National AI Strategy 2020–2045 (BPPT, 2020); BOPTN allocation cut by about 50% in budget efficiency (kumparanNEWS, 2025)	A zero-licence, policy-aligned model is attractive to government and public universities
Economic	Tight university budgets and limited ability to pay for additional licences; price-sensitive emerging-market users	Free institutional distribution; paid conversion treated as upside only

Social	51% of students worry about critical thinking (Agusnaya & Nirmala, 2025); usage outpaces training by 39.7 points (Oktaviani et al., 2026)	Educator training and process evidence are as important as the product
Technological	Internet penetration of 83.64% in Java versus 67.6% in 3T areas (Septiani, 2024)	Low-bandwidth mode, cached sources and zero-rating
Environmental	Cloud-delivered service; on-campus footprint limited to caching kits for about 120 campuses	Low materiality; specify energy-efficient caching hardware
Legal	Personal Data Protection Law No. 27/2022 (Republic of Indonesia, 2022); UNESCO (2023) recommends mandated data-privacy protection	Privacy Impact Assessment in Year 1; aggregated, consent-based indicators only

4. Proposed Solution

4.1 Concept: adaptive friction

SOCRATIC AI is not a restriction layer. It follows a principle of **adaptive friction**: where the goal is learning, Gemini asks, challenges and scaffolds first; where the goal is productivity (formatting, translation, scheduling), it simply helps. Students always retain control through a **hint ladder** (question, hint, worked step, full solution plus a reflection prompt). This preserves autonomy and lowers the risk that students abandon Gemini for friction-free alternatives.

4.2 The 3C Socratic Framework

Table 6. The 3C framework: experience, mechanism and outcome

Pillar	What the student experiences	Product mechanism	Learning outcome	Segments
Challenge	Gemini acts as a devil's advocate on drafts: it surfaces unexamined assumptions, logical fallacies and the strongest counterargument, and asks what evidence would change the student's mind, without rewriting the draft	Challenge template in Notebook; Socratic mode in the Gemini app	Argument quality, intellectual humility	A, C
Check	Every synthesized claim is anchored to uploaded lecture slides, journals or syllabi, with inline citations and page references; unsupported claims are flagged as "not found in sources"	Source-grounded Notebook; evidence map; citation checklist	Verification habit, academic integrity	A, B
Construct	Guided problem-solving in Bahasa Indonesia: hypothesis, data interpretation, self-explanation and reflection, with graded hints	Prompt library; Guided Notebooks; Learning Journal export to Docs	Conceptual understanding, retention	C, B

4.3 Localization and feature enhancements

- **Socratic Prompt Library (Bahasa Indonesia).** More than 60 templates across six discipline clusters (engineering and IT, health, natural science, social science and humanities, economics and business, education), mapped to the 3C pillars (samples in Appendix A).

- **Course-ready Notebook templates.** Literature Review, Thesis Proposal (skripsi), Exam Preparation and Lab Report notebooks that follow Indonesian academic conventions and encourage use of Indonesian scholarly sources.
- **Classroom integration.** A lecturer assigns a Guided Notebook with course sources and Socratic mode switched on; students submit a Learning Journal as process evidence. Lecturers see consent-based, aggregated process indicators, not private chat logs.
- **Low-bandwidth mode.** Text-first interface, cached course sources and deferred synchronization for unstable connections in 3T regions.
- **Educator toolkit.** Assessment-redesign guides (oral defence, process portfolios, source-grounded assignments) and an AI-use policy template aligned with the National AI Strategy 2020–2045.

Table 7. Use-case map by segment

Segment	Priority use cases	Primary tool	3C emphasis
A Deep Researchers	Multi-paper literature synthesis with citation checks; research-gap identification; critique of thesis outlines	Gemini Notebook	Check, Challenge
B Practical Innovators	Data analysis and Python on lab or business datasets; financial modelling; workflow prototyping with explain-your-code reviews	Gemini app	Construct, Check
C Foundational Learners	Concept clarification with local examples; step-by-step problem solving; "quiz me" exam preparation; study planning	Gemini app and Starter Pack	Construct, Challenge

Gemini already includes learning-oriented capabilities. Our contribution is therefore the *Indonesian localization, institutional workflow and measurement layer* around them. We recommend Google validate overlap with existing guided-learning features before the Year-1 build scope is fixed.

4.4 How it works in practice

A second-year student at a public university in an outer region receives a Classroom assignment. (1) She opens the lecturer's Guided Notebook, which contains the course readings. (2) Instead of asking for the answer, she writes her hypothesis, and Gemini challenges one assumption and asks for supporting evidence. (3) Notebook answers every question with page-referenced citations, and flags a claim that the readings do not support. (4) On a weak connection the low-bandwidth mode keeps working from cached sources. (5) She exports a Learning Journal and submits it, giving her lecturer evidence of her reasoning, and herself protection against unfair suspicion of misuse.

4.5 Responsible AI by design

Privacy: the programme runs on Workspace for Education accounts, where data is not human-reviewed or used to train models (Google Workspace Admin Help, n.d.), and complies with Indonesia's Personal Data Protection Law (Republic of Indonesia, 2022). **No surveillance:** lecturers receive aggregated, consent-based indicators only, addressing the surveillance and transparency concerns in the literature (Susanto et al., 2026). **Human in the loop:** educators, not the model, decide assessment outcomes. **Accuracy:** grounding and citations reduce, but do

not eliminate, error, so students are taught to verify. **Accessibility:** templates are tested with students using assistive technologies and on low-end devices.

4.6 Pedagogical foundation of the 3C framework

The 3C design is not an arbitrary interface choice; each mechanism is tied to established learning science and to the recent evidence in Section 2.5 (Table 6a).

Table 6a. Learning-science principles and the matching design response

Principle	Evidence base	Design response in SOCRATIC AI
Scaffolding within the zone of proximal development	Vygotsky (1978)	The hint ladder moves from a question to a hint, a worked step and finally a full solution, so support fades as the student progresses.
Tutor-like, personalized guidance at scale	Bloom (1984); Kestin et al. (2025)	Gemini provides one-to-one style questioning that lecturers cannot give to every student, while lecturers keep class time for higher-order work.
Retrieval practice (testing effect)	Roediger & Karpicke (2006)	"Quiz me" exam preparation and Construct prompts ask students to recall and explain before the model reveals content.
Desirable difficulties	Bjork & Bjork (2011)	Adaptive friction applies challenge only where learning is the goal; a one-click "just answer" with a reflection prompt protects student autonomy.
Guardrailed AI design	Bastani et al. (2025)	Prompts hold back final answers, give feedback on drafts without rewriting them, and produce a Learning Journal as proof of process.

Design caution. Excess friction can push students towards friction-free tools, which is why friction is adaptive and student-controlled and why drop-offs after Socratic mode are an early-warning indicator in the risk register (Table 12).

5. Go-to-Market Strategy

5.1 Three-tier model and adoption funnel

The go-to-market plan has three tiers. **Top-down:** partner with Kemdiktisaintek and public universities to enable Gemini Notebook inside existing Workspace for Education domains, at no additional licence cost. **Enabler:** partner with telecommunications providers (for example Telkomsel and Indosat) for zero-rated access to Gemini educational domains, combined with the low-bandwidth mode. **Bottom-up:** the **#ThinkWithGemini** campaign, powered by 500 Google Student Champions in student organizations (BEM, academic clubs, local networks). Positioning: **"Think with Gemini, not instead of it."** The funnel follows the AARRR model (Figure 2, Table 8), consistent with the case's awareness, trial, adoption and sustained-usage chain.

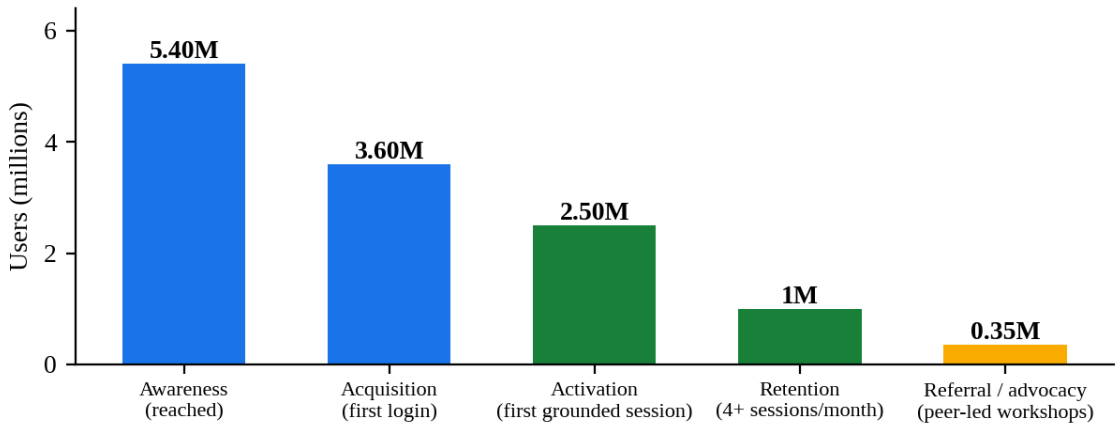


Figure 2. Year-3 base-case adoption funnel (millions of users)

Source: team model; stage ratios are planning assumptions (Appendix B).

Table 8. AARRR funnel plan

Stage	Key tactics	Owner / channel	Year-3 KPI
Awareness	#ThinkWithGemini during orientation weeks (PKKMB); short-form student-made content; "Think First" challenge	Champions; student organizations; social media	5.4M reached
Acquisition	Admin enablement kit (checklist, 30-minute webinar); campus launch events; Classroom announcements	Google for Education team; IT units	3.6M first login
Activation	"First Socratic Session in 7 days" onboarding; Starter Pack of 10 prompts	In-product onboarding; lecturers	2.5M activated (≥1 session/month)
Retention	Seasonal packs (exam prep, thesis); lecturer-assigned Guided Notebooks; Learning Journal	Lecturers; Classroom	1.0M sustained (4+ sessions/month)
Referral	Peer workshops on ethical research; hackathons; champion certification	Champions; student organizations	350k reached through peers

5.2 Campaign rhythm and partnerships

Campaigns follow the academic calendar: **orientation weeks** (introduce "Think First" to new students), **mid-terms** (Exam Preparation Notebooks), **finals** (research hackathons and case competitions), and **thesis season** (Literature Review and Thesis Proposal Notebooks). Distribution is institutional rather than paid-media-led, so acquisition cost is mainly champion stipends and training. The partnership plan is:

- **Government:** Kemdiktisaintek co-develops the governance guideline and endorses pilots; the National Human-Centered AI Learning Index is jointly branded.
- **Universities:** 50 pilot institutions in Year 1, at least 15 of them outside Java, expanding to 150 in Year 2 and 300+ in Year 3.
- **Telcos:** zero-rated educational domains with shared usage dashboards; their return is education-linked traffic and brand association.
- **Student organizations:** BEM and academic clubs host workshops and hackathons, giving champions a ready peer network.

For brand impact, we track lecturer recommendation (NPS), unaided association of Gemini with "AI for learning", and the share of partner universities citing the Socratic Framework in official guidelines.

5.3 Brand impact framework and stakeholder messaging

Because the competition objective is brand impact, we translate brand health into measurable stages instead of relying on usage counts alone (Table 8a). Targets are team planning targets and will be baselined in the first semester.

Table 8a. Brand-impact ladder: from awareness to institutional endorsement

Brand stage	Question answered	Metric	Instrument	Year-3 target
Awareness	Do students and lecturers know Gemini has a learning-first mode?	Aided awareness of SOCRATIC AI / #ThinkWithGemini on partner campuses	Annual survey	Baseline in Year 1; +20 pp by Year 3
Association	Is Gemini linked to "AI for learning"?	Unaided association of Gemini with "AI for learning"	Annual survey	+15 pp versus Year-1 baseline
Trust	Do lecturers recommend it?	Lecturer NPS for Gemini as a learning tool	Semester pulse survey	≥ +40 (Table 13)
Endorsement	Do universities adopt the framework?	Share of partner universities citing the Socratic Framework in official guidelines	Policy audit	≥50% of partners
Loyalty	Do students keep using it habitually?	Sustained users (4+ sessions/month); eight-week retention	Aggregated telemetry	1.0M sustained users

Table 8b. Message architecture by audience

Audience	Core message	Proof point
Students	"Think with Gemini, not instead of it."	Learning Journal protects them against unfair suspicion of AI misuse
Lecturers	"See the thinking, not just the answer."	Aggregated process indicators and ready-made assessment kits
University leaders	"Governance-ready AI at no extra licence cost."	Workspace for Education data protection; PDP Law-aligned governance kit
Government	"Human-centred AI that reaches every campus."	Ring-fenced 3T budget; National Human-Centered AI Learning Index
Telcos	"Education traffic with a measurable social return."	Shared usage dashboards on zero-rated educational domains

5.4 Lecturer adoption playbook

Faculty resistance is rated a medium-likelihood, high-impact risk (Table 12), so lecturers are treated as the primary channel rather than a side audience. We use a three-step ladder. **Aware:** a 30-minute enablement webinar and a one-page explanation of what Socratic mode does and does not do. **Try:** one ready-to-run Guided Notebook assignment per course type, designed to take under 30 minutes to set up. **Adopt:** co-designed assessment redesign, a peer-lecturer community and a certificate. We will also explore with Kemdiktisaintek whether the certificate can be recognised in lecturers' workload reporting; this is to be validated in Year 1. Progress is tracked by the share of trained educators who assign a Guided Notebook (gate: 70% in Year 2, Table 9).

6. Implementation Roadmap, Financials and Risk

6.1 Three-year roadmap with phase gates

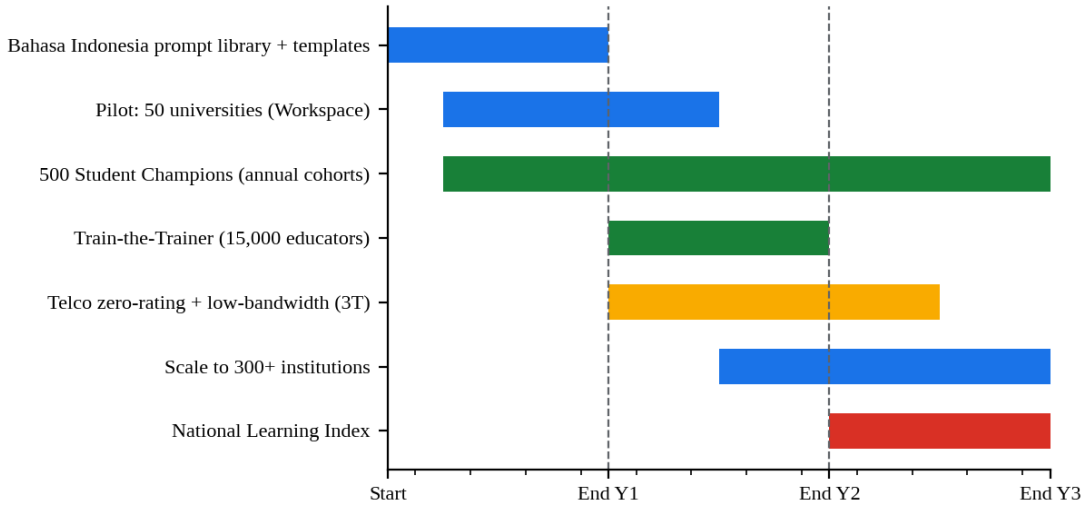


Figure 3. Implementation timeline by quarter

Table 9. Phases, targets and decision gates

Phase	Key milestones	Targets	Gate to next phase
Year 1 Foundation and zero-cost adoption	Deploy Bahasa Indonesia prompt library and Notebook templates; pilot in 50 universities; certify 500 Champions; baseline survey; privacy impact assessment	50 institutions; 0.3M activated users	≥40% grounded sessions; lecturer NPS ≥ +30
Year 2 3T expansion and educator upskilling	Telco zero-rating; low-bandwidth mode; Train-the-Trainer for 15,000 educators; Classroom Guided Notebook launch	150 institutions; 1.2M activated; ≥12% from 3T	≥50% grounded sessions; 70% of trained educators assign Guided Notebooks
Year 3 National scaling and policy integration	Expand to 300+ institutions; publish the National Human-Centered AI Learning Index; hand over to co-funded operating model	300+ institutions; 2.5M activated; ≥15% from 3T	Index published; sustainability plan signed

Year-1 pilot design. To validate assumptions before scaling, the 50 pilot campuses are matched into three arms: (1) standard Gemini access (control), (2) access plus the Socratic Prompt Library and student training, and (3) arm 2 plus lecturer-assigned Guided Notebooks and educator training. We compare grounded-session rate, eight-week retention, a critical-thinking pulse score and lecturer NPS. The results decide the Year-2 scope and the Year-1 gate.

6.2 Budget and resource requirements

The fund totals **US$1.5 million** over three years, phased to match build effort in Year 1, field deployment in Year 2 and consolidation in Year 3 (Table 10, Figure 4). Each line includes roughly 5% contingency. The fund covers programme spend; the core programme office (about eight FTE across product localization, partnerships, research and operations) is assumed to be provided by Google and its implementation partner outside the fund.

Table 10. Budget by category and year

Category	%	Y1	Y2	Y3	Total	Main line items (US$ thousand)
Infrastructure and regional access	30%	60	240	150	450	Campus caching and connectivity kits for about 120 3T campuses (300); telco zero-rating co-funding (150)
Educator and Champion upskilling	27%	105	200	100	405	Champion programme (100); Train-the-Trainer for 15,000 educators, about US$15 each through a cascade model (225); certification and content (80)
Product localization and academic UI	23%	195	90	60	345	Prompt library and templates (150); Classroom and low-bandwidth UX prototyping (120); QA and accessibility testing (75)
Community campaigns and hackathons	20%	90	100	110	300	#ThinkWithGemini campaign (140); hackathons and case competitions (110); independent evaluation (50)
Total	**100%**	**450**	**630**	**420**	**1,500**	

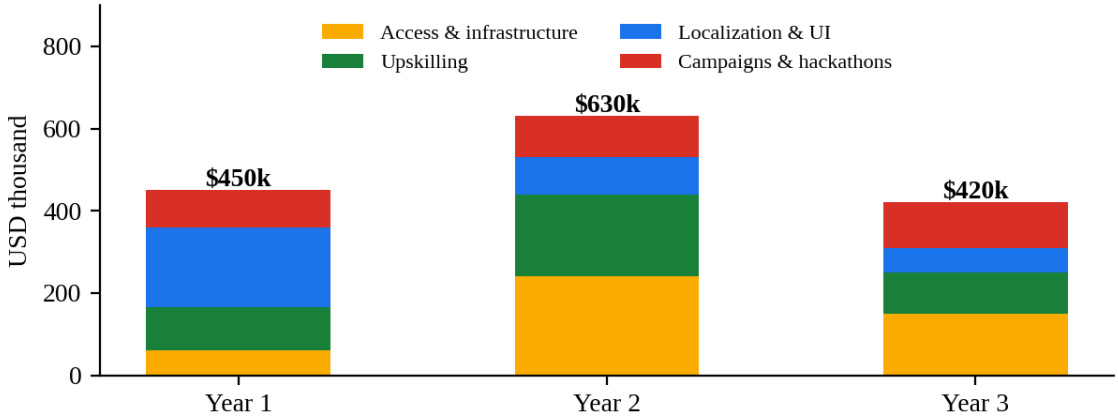


Figure 4. Annual budget phasing by category (US$ thousand)

6.3 ROI and unit economics

Financial return is modelled conservatively and treated as a *lower bound*, because it excludes brand value, Workspace retention and institutional cost avoidance. The model assumes that a small share of activated students later converts to a paid Google AI plan at a blended price of US$36 per user-year (an emerging-market assumption) with a 70% contribution margin (Table 11).

Table 11. Scenario analysis (planning assumptions, to be validated in Year 1)

Metric (Year 3)	Conservative	Base	Optimistic
Activated users	1.2M	2.5M	3.5M
Sustained users (4+ sessions/month)	0.45M	1.0M	1.5M
Programme cost per activated user	US$1.25	US$0.60	US$0.43
Paid conversion (illustrative)	1.0%	1.5%	2.0%
Paying users	12,000	37,500	70,000
Annual contribution run-rate	US$0.30M	US$0.95M	US$1.76M
Payback after conversions begin	about 60 months	about 19 months	about 10 months

Break-even requires about 59,500 paid user-years (US$1.5M divided by US$25.2 contribution per user-year). Impact-based unit costs are favourable: about **US$15 per trained educator**, and about **US$1.20 per 3T user** in the base case (US$450,000 of access spend over 375,000 users). Because the programme is justified primarily by adoption depth, brand trust and equity, Year-2 and Year-3 gates, not the financial payback alone, govern release of funds.

Sensitivity. Payback is most sensitive to paid conversion: halving it doubles payback from about 19 to 38 months. A change of ±20% in activated users moves the cost per activated user between US$0.50 and US$0.75. The Year-1 pilot is designed to narrow both uncertainties before the larger Year-2 spend is released.

6.4 Governance and operating model

A **Steering Committee** (Google for Education Indonesia, Kemdiktisaintek, three university representatives and a telco representative) meets each semester and decides on phase gates. A **programme office** coordinates four workstreams: product localization, institutional partnerships, educator and Champion programmes, and research and evaluation. Regional coordinators drawn from the Champion network support campuses outside Java.

6.5 Risks and mitigation

Table 12. Risk register (L = likelihood, I = impact; H = high, M = medium, L = low)

Risk	L	I	Mitigation	Early-warning indicator
Excess friction drives students to other tools	M	H	Adaptive friction and hint ladder; A/B tests; one-click "just answer" with reflection	Drop in weekly active use after Socratic mode
Privacy or surveillance concerns	L	H	Aggregated, consent-based indicators only; Privacy Impact Assessment; PDP Law compliance	Complaints; admin opt-outs
Telco partnership delays	M	M	Campus caching and low-bandwidth mode work without telco; phased rollout	No MoU by month 12
Faculty resistance or limited capacity	M	H	Train-the-Trainer cascade; Classroom-ready assessment kits; certificates	Low assignment of Guided Notebooks
Overlap with existing Gemini features	M	M	Early product review; position as localization and institutional layer	Duplicate feature flags in review
Equity gap widens	M	H	Ring-fenced 30% access budget; 3T share tracked each semester	3T share below gate
Policy or funding shifts	M	M	Zero-licence design; multi-partner co-funding	BOPTN or Workspace policy changes

6.6 Long-term sustainability

After Year 3, the run-rate falls to about **US$300,000–350,000 per year** (library maintenance, one Champion cohort, evaluation and hosting of the Index). It is financed by a mix of Google for Education's regional budget, university co-funding and telco CSR contributions. Sustainability also comes from structure: the Train-the-Trainer cascade and alumni Champions keep capability on campus, while the prompt library is published as an open educational resource that lecturers can adapt.

7. Expected Impact and Measurement

Impact is measured at three levels: adoption (are students using it?), behaviour (are they using it to think?) and outcomes (is learning and trust improving?). Table 13 sets targets; baselines are established in the first semester.

Table 13. Key performance indicators

KPI	Y1	Y2	Y3	Measurement
Partner institutions integrated	50	150	300+	Admin enablement records
Activated users (≥1 session/month)	0.3M	1.2M	2.5M	Aggregated product telemetry
Sustained users (4+ sessions/month)	0.1M	0.5M	1.0M	Aggregated product telemetry
Share of activated users from 3T regions	≥8%	≥12%	≥15%	Institution coding
Notebook sessions with source grounding	≥40%	≥50%	>60%	Session-level flags
Educators trained (cumulative)	1,500	15,000	15,000+	Certification records
Students concerned that AI harms critical thinking	Baseline 51%	≤45%	≤35%	Annual survey (baseline: Agusnaya & Nirmala, 2025)
Partner institutions with formal AI-use guideline	50%	80%	100%	Policy audit
Lecturer NPS for Gemini as a learning tool	≥ +30	≥ +35	≥ +40	Semester pulse survey

Data is collected as anonymized, aggregated telemetry under user consent, complemented by pre/post pulse surveys and an independent evaluator (US$50,000 in the budget). The 3T share is an equity-weighted target and will be recalibrated against enrolment data in the baseline semester. The wider impact is as follows: **students** gain critical-thinking and research skills; **educators** receive training and credible evidence of student work; **institutions and government** obtain a zero-licence, policy-aligned model and a national benchmark; **Google** earns durable, trust-based usage.

8. Conclusion

The question is not whether Indonesian students will use AI, but whether it will make them think more or less. SOCRATIC AI answers by changing the design of the interaction, equipping the educators who guide it, and removing the cost and connectivity barriers that would otherwise limit it to a privileged few. For approximately US$0.60 per activated user, Google can turn Gemini and Gemini Notebook from a task shortcut into a trusted learning companion and, in doing so, build the most durable form of brand impact in education: trust earned one lecture hall at a time.

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Appendix A. AI Generated Infographic

Image A1. Infographic

